

Institutional Response to Public Testimony on Redistricting Communities of Interest

Spencer Kuhn

The GeoNames database compiles place names alongside their corresponding point or polygon geometries at all administrative levels around the world (“GeoNames Gazetteer Extract Files” 2026). Pennsylvania GeoNames locations serve as the building block unit of each commenter’s community of interest.

Public testimony data was gathered from the Redistricting Data Hub, which has compiled comments from the 2020 redistricting cycle for 14 individual states (Redistricting Data Hub 2022). Tabula, a software capable of converting PDFs to csv files, was used to process comments about Pennsylvania state house boundaries and commenter communities of interest.

The `ellmer` package in R was used to interact with off-the-shelf LLM chat features and extract structured data (Wickham et al. 2025), including comment sentiment, comment clarity, and every specific location mentioned within a comment. Comment sentiment and clarity are both scaled to range from 0 to 1, with 0 representing negative sentiment and unclear comments while 1 represents positive sentiment and clear comments. The date of each comment was available from the original dataset. The `ellmer` package was then used to estimate the complete set of Pennsylvania counties contained within each commenter’s mentioned locations.

For each comment, potential GeoNames entries were filtered to the corresponding set of identified counties, and the filtered list was matched using the `stringdist` package in R to the comment’s specific location mentions (Van Der Loo 2014). Lastly, `ellmer` was again used to group these locations into distinct communities of interest based on the commenter’s specifications.

“GeoNames Gazetteer Extract Files.” 2026. GeoNames. <https://www.geonames.org/>.

Redistricting Data Hub. 2022. “Public Redistricting Testimony.” Harvard Dataverse. <https://doi.org/10.7910/DVN/11CKIN>.

Van Der Loo, M. P. J. 2014. “The Stringdist Package for Approximate String Matching.” *The R Journal* 6 (1): 111–22. <https://doi.org/https://journal.r-project.org/articles/RJ-2014-011/>.

Wickham, Hadley, Joe Cheng, Aaron Jacobs, Garrick Aden-Buie, and Barret Schloerke. 2025. *Ellmer: Chat with Large Language Models*. <https://doi.org/10.32614/CRAN.package.ellmer>.